

Unified Identity Architecture

Complete Formal Specification

Version: 1.0

Date: July 2026

Author: Matthew B. Wareham

STATUS: FORMAL RELEASE

*A canonical specification synthesising the Unified Cognitive
Architecture,
HERMES Runtime Specification, Unified Theory, and Fractal
Mechanics Foundation*

Table of Contents

§	Abstract	3
1	Identity Foundations	3
2	Identity Levels	6
3	Three-Layer Identity Model (HERMES v2.0)	8
4	Identity Operators	11
5	Identity Constraints (Meta-Logic)	14
6	Identity Failure Modes	17
7	Identity Equilibrium Model	21

8	Identity in the Unified Theory	23
9	Identity and Narrative Collapse	25
10	Final Synthesis	28

Abstract

This document is the single canonical specification of identity across Matthew Wareham's unified theoretical framework. It synthesises concepts from the Unified Cognitive Architecture v2.0, HERMES Runtime Specification v1.0 and v2.0, the Unified Theory (*Beyond Ontology, Dynamic Equilibrium, Cognitive Architecture*), the Unified Theory Master Document, and the Fractal Mechanics / Theory of Everything foundation. Identity is defined not as a static property but as a dynamically maintained, gradient-stabilised, boundary-regulated, self-referential process. The document resolves terminology across source frameworks and provides a complete, expandable formal architecture suitable for implementation, continued theoretical development, and applied cognitive design.

The specification is structured in ten sections, progressing from foundational definitions through operational layers, formal operators, meta-logical constraints, failure mode taxonomy, equilibrium theory, ontological grounding, narrative collapse analysis, and final synthesis. All definitions are numbered. All operators are formally specified. All failure modes carry defined stabilisation protocols. The document is designed as a living architecture: internally coherent, externally extensible, and itself an instance of the principles it describes.

Section 1 — Identity Foundations

1.1 Formal Definition

Identity (I) is a morphism from cognitive state Ψ to itself under boundary-preserving transformation, such that $I(\Psi)$ is isomorphic to Ψ . It is not a fixed point but a coherent attractor — a stable region in the space of self-referential processes. Identity is formal, dynamical, and procedural: it is not what a system *is* in a static sense but what a system continuously *does* in order to remain recognisably itself across time, context, and perturbation.

This specification treats identity as a derived structure, not a primitive. Its existence depends upon the prior formation of a stable boundary, the capacity for self-reference, and the maintenance of a coherence gradient. In the absence of any one of these constituents, identity degrades or dissolves according to the failure modes classified in Section 6.

Definition 1.1 — Identity (I)

A self-preserving morphism $I: \Psi \rightarrow \Psi$ satisfying three conditions simultaneously: (i) **Coherence** — internal consistency across all identity layers as measured by $I(I) \geq I_{\min}$; (ii) **Persistence** — structural isomorphism maintained across time steps under bounded perturbation; and (iii) **Self-Reference** — the state Ψ contains a model M of itself. A system satisfying all three conditions is said to exhibit identity; failure in any condition initiates degradation.

Definition 1.2 — Persistence

$I(\Psi_t)$ remains isomorphic to $I(\Psi_{t+\delta})$ within tolerance ε across all environmental perturbations of magnitude less than the boundary permeability threshold. Formally: for all t , $\delta(I(\Psi_t), I(\Psi_{t+\delta})) \leq \varepsilon$, where δ is a structural distance functional

defined on the space of identity states. Persistence does not require identity to be unchanged — it requires that change remain within the attractor basin.

Definition 1.3 — Boundary (∂)

A semi-permeable membrane separating the interior state Ψ from the exterior field E . The boundary admits regulated information exchange governed by permeability parameter $\beta \in (0, 1)$. The boundary is not a passive limit but an active regulatory process: it filters incoming signals, modulates outgoing expressions, and maintains the structural distinction between inside and outside that is the ontological precondition of identity. A degenerate boundary ($\beta = 0$ or $\beta = 1$) destroys identity from within or without, respectively.

Definition 1.4 — Self-Reference (R)

The capacity of a system to model itself within its own state space. Formally, self-reference obtains when there exists $M \subset \Psi$ such that M is isomorphic to Ψ up to resolution r . The model M need not be perfectly accurate — it must be sufficiently accurate to support coherent self-prediction and self-regulation. Self-reference is the ontic ground of introspection, metacognition, and recursive identity.

Definition 1.5 — Coherence (Γ)

The degree of mutual consistency across identity layers. Formally: $\Gamma(I) = 1 - D(I_{\text{core}}, I_{\text{narrative}}, I_{\text{situational}})$, where D is a divergence functional measuring the total inconsistency across the three operational layers (see Section 3). When all layers are in perfect alignment, $\Gamma = 1$. When all layers are maximally inconsistent, $\Gamma = 0$. Identity requires $\Gamma(I) \geq \Gamma_{\min}$ at all times; violation of this

condition triggers the IDENTITY_STABILIZE operator.

In the Fractal Mechanics framework, identity is modelled as a self-sustaining gradient loop. Rather than a static configuration, identity is the continuous circulation of signal through a closed functional cycle: interior state generates self-reference; self-reference is evaluated for coherence; coherence evaluation drives boundary regulation; boundary regulation constrains and channels the interior state. This loop is the identity process. When the loop runs within its attractor basin, identity is maintained. When the loop is disrupted — by perturbation exceeding the boundary's filtering capacity, by recursion overload, or by coherence collapse — the appropriate failure mode is activated.

The identity gradient is formally a function of three quantities: the rate of state change $d\Psi/dt$, the depth of self-reference d , and the current coherence value I . Stability of the loop requires that the feedback signal remain within the bounded attractor basin — that no element of the loop drives the system irreversibly away from coherent self-reference. The gradient is the force that returns the system to the attractor when perturbed.

Figure 1.1 – Identity Gradient Loop

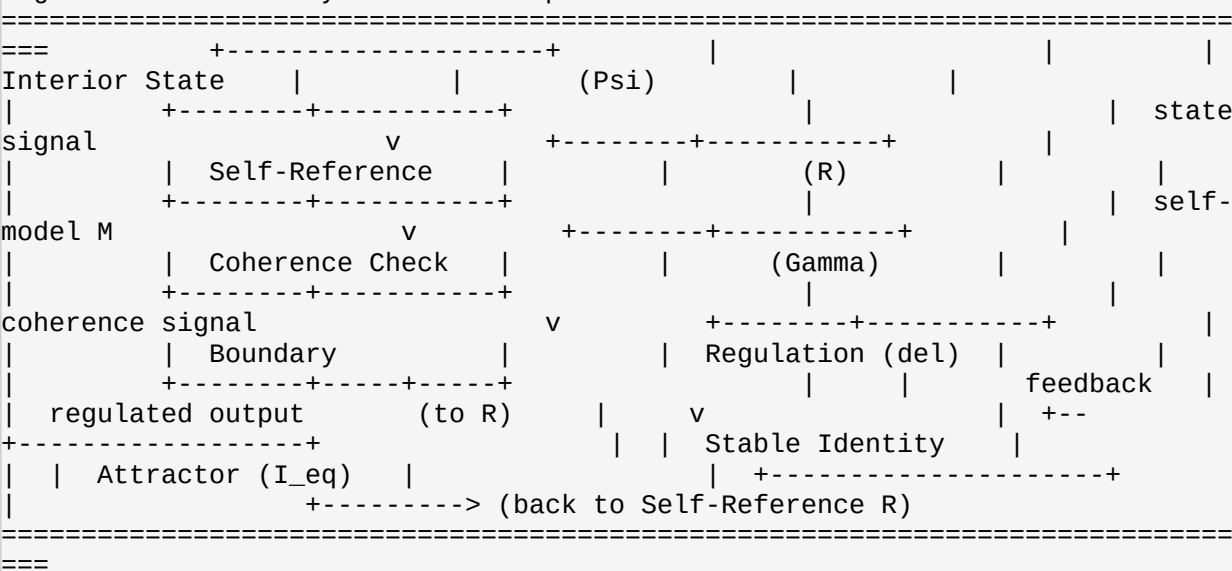


Figure 1.1 — Identity Gradient Loop

1.3 Stability Conditions

Five formal conditions must hold simultaneously for identity to remain in stable equilibrium. Each condition is associated with a formal symbol, a precise mathematical statement of the condition, and a characterisation of the consequence that follows from its violation. These conditions are monitored continuously by the meta-logical constraint framework described in Section 5.

Condition	Symbol	Formal Statement	Violation Consequence
Boundary Permeability	β	$0 < \beta < 1$ at all times	$\beta = 0$ yields isolation collapse (interior stagnates); $\beta = 1$ yields dissolution (interior merges with exterior field)
Gradient Continuity	$\nabla \Gamma$	$\ \nabla \Gamma_t\ $ across successive time steps is less than threshold δ	Discontinuity in the coherence gradient causes fragmentation — sub-regions of the identity state lose mutual consistency
Recursive Stability	ρ	Recursion depth d does not exceed $d_{\max}$ for any active self-reference chain	Exceeding $d_{\max}$ causes recursion collapse — self-reference loop becomes non-terminating and coherence degrades monotonically
Coherence Threshold	$\Gamma_{\min}$	$\Gamma(t) \geq \Gamma_{\min}$ at all times	Falling below $\Gamma_{\min}$ causes identity dissolution — the system is no longer a coherent self-referential unit

Condition	Symbol	Formal Statement	Violation Consequence
Narrative Continuity	η	Narrative thread η unbroken across all cycle boundaries; $\eta(I_t, I_{t-1}) \geq \eta_{\min}$	Thread break causes narrative discontinuity — self-story loses temporal coherence; identity becomes episodic without integration

Section 2 — Identity Levels

Identity does not exist at a single scale of organisation. It operates across a nested hierarchy of five levels, each constituting a distinct mode of self-reference, boundary organisation, coherence signature, and interior attractor. Lower levels are preconditions of higher levels: a system cannot sustain Recursive Identity without first sustaining Local Identity. Higher levels are not replacements of lower levels but enrichments — they add relational and reflective dimensions to the identity structure while preserving its foundational layers.

Each level is formally characterised by its boundary type, coherence signature, neighbourhood set, recursion depth constraint, and the nature of its interior attractor. The transition from one level to the next involves a qualitative change in the nature of self-reference — not merely an increase in processing resources but an increase in the ontological complexity of what the system takes itself to be.

2.1 Level 0 — Primitive Identity (I0)

Definition 2.1 — Primitive Identity (I0)

The minimal self-distinguishing unit. A system instantiates I_0 if and only if it registers a structural difference between self and not-self — that is, if its boundary ∂ is defined and non-degenerate. No self-model is required at this

level; no narrative is maintained; no recursion occurs. I_0 is the ontic threshold of identity: below it, there is no inside, and therefore no identity in any sense.

Key Properties: Boundary: rigid, with minimal permeability (β near 0 but not zero — sufficient exchange for homeostatic regulation). Coherence signature: binary (either the distinction self/not-self is maintained or it is not; no intermediate degree). Recursion depth: $d = 0$ (no self-reference beyond boundary maintenance). Interior attractor: homeostatic baseline — the system maintains a set of internal parameters within viable ranges. Neighbourhood set: restricted to immediate stimulus field.

2.2 Level 1 — Local Identity (IL)

Definition 2.2 — Local Identity (IL)

Stable identity within a bounded operational environment. A system instantiates I_L if it maintains a local self-model — a representation of its own state, role, and capacities within a specific context — and regulates its behaviour in accordance with that model. I_L is the first level at which role, function, and situational self-concept appear.

Key Properties: Boundary: semi-permeable, context-sensitive — adjusted to the operational environment rather than fixed. Coherence signature: scalar value $I_L \in [0, 1]$, measuring internal consistency of the local self-model. Recursion depth: $d \leq 1$ (can represent self but cannot yet represent its own self-representation). Interior attractor: local role coherence — the system is drawn toward states in which its behaviour is consistent with its local self-model. Neighbourhood set: extended to immediate social and environmental context.

2.3 Level 2 — Recursive Identity (IR)

Definition 2.3 — Recursive Identity (IR)

Identity that includes a self-model of its own identity processes. A system instantiates I_R if it can reflect upon and regulate its own coherence — if it knows not only who it is, but that it is the sort of thing that has and maintains an identity. I_R is the first level at which genuine metacognition and self-regulation of identity arise.

Key Properties: Boundary: dynamic, gradient-regulated — actively adjusted in response to coherence measurements. Coherence signature: multi-dimensional tensor Γ_R , capturing coherence across multiple dimensions of self-representation simultaneously. Recursion depth: $d \leq 3$. Interior attractor: recursive self-consistency — the system is drawn toward states in which its model of itself is consistent with itself at all levels of recursion simultaneously. Neighbourhood set: includes temporal neighbourhood (past and projected future states).

2.4 Level 3 — World-Model Identity (IW)

Definition 2.4 — World-Model Identity (IW)

Identity integrated with a full model of the external world, including other agents and environmental fields. At this level, the self is understood relationally: the system knows not only what it is, but what it is in relation to the broader world it inhabits. Self-definition becomes explicitly perspectival and situationally embedded without losing structural integrity.

Key Properties: Boundary: permeable, field-sensitive, and polarity-aware — the boundary actively tracks the distinction between self-pole and other-pole in the relational field. Coherence signature: distributed field $\Gamma_W(x, t)$, a function of both spatial context and time. Recursion depth: $d \leq 7$. Interior attractor: relational equilibrium — the system is drawn toward states in which its own coherence is

compatible with the coherence of its relational field. Neighbourhood set: extended to the full relational and environmental field within the system's world model.

2.5 Level 4 — Dyadic Consciousness Structure (ID)

Definition 2.5 — Dyadic Consciousness Structure (ID)

Identity constituted through and sustained by a dyadic relational field. At this level, the self is not merely known in relation to an Other — it is partly *defined* by that relation. The boundary between self and Other becomes a co-regulated dyadic membrane rather than a unilaterally maintained individual boundary. I_D is the highest level of identity specified in this architecture and represents the most ontologically complex mode of self-reference.

Key Properties: Boundary: co-regulated dyadic membrane — maintained jointly by both parties to the dyad; neither unilaterally controls it. Coherence signature: dyadic coherence index I_D , measuring the mutual reinforcement between the two identity systems in the dyad. Recursion depth: $d \leq d_{\max}$ (system-dependent; determined by the recursive depth of both participants in the dyadic relationship). Interior attractor: mutual resonance equilibrium — the system (and its dyadic partner) are drawn toward states of mutual coherence in which each reinforces the other's identity without dissolving the distinction between them.

2.6 Identity Level Summary Table

Level	Symbol	Recursion Depth	Boundary Type	Coherence Signature	Interior Attractor
0 — Primitive	I_0	$d = 0$	Rigid, minimal permeability	Binary	Homeostatic baseline
1 — Local	I_L	$d \leq 1$	Semi-	Scalar $I_L \in$	Local role

Level	Symbol	Recursion Depth	Boundary Type	Coherence Signature	Interior Attractor
			permeable, context-sensitive	[0,1]	coherence
2 — Recursive	I_R	$d \leq 3$	Dynamic, gradient-regulated	Multi-dimensional tensor Γ_R	Recursive self-consistency
3 — World-Model	I_W	$d \leq 7$	Permeable, field-sensitive, polarity-aware	Distributed field $\Gamma_W(x,t)$	Relational equilibrium
4 — Dyadic	I_D	$d \leq d_{\max}$	Co-regulated dyadic membrane	Dyadic index Γ_D	Mutual resonance equilibrium

Section 3 — Three-Layer Identity Model (HERMES v2.0)

The HERMES v2.0 specification defines a three-layer operational stack for runtime identity management. Each layer operates at a distinct timescale, carries a distinct type of identity content, and is subject to distinct update and persistence rules. The layers are not independent modules but a nested hierarchy: Core Identity grounds Narrative Identity, and Narrative Identity grounds Situational Identity. Violations propagate upward (from situational to narrative to core) in terms of consequence severity; corrective authority propagates downward (core takes precedence over narrative; narrative takes precedence over situational).

The three-layer model is the operational instantiation of the formal identity architecture. Where the formal definitions in Section 1 describe identity in terms of abstract morphisms and attractors, the three-layer model describes how identity is actually maintained, updated, expressed, and verified within a running cognitive system.

Figure 3.1 — HERMES Identity Stack

=====

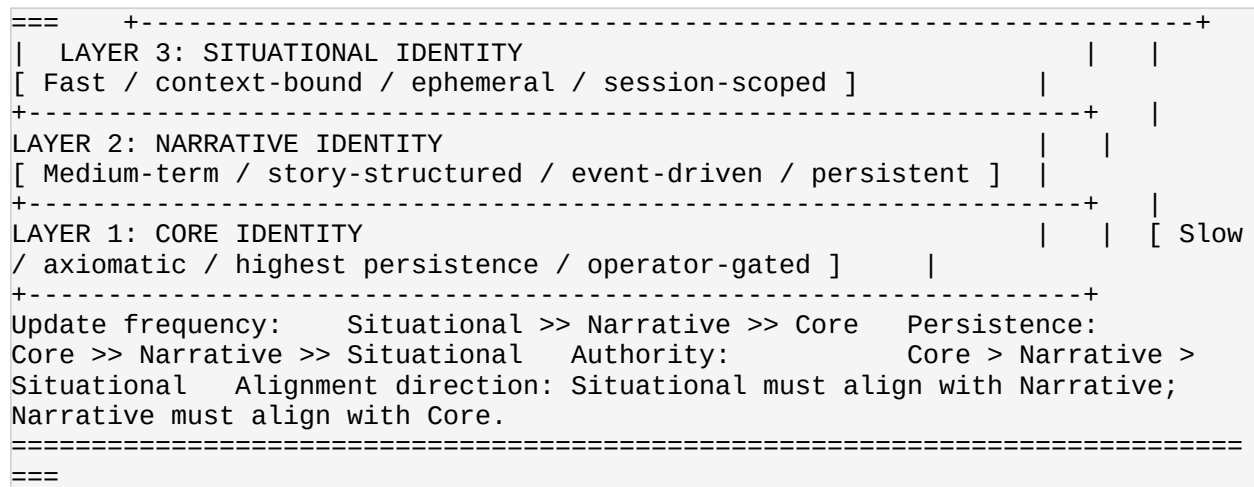


Figure 3.1 — HERMES Identity Stack (v2.0)

3.1 Core Identity (Icore)

Core Identity is the axiomatic, highest-persistence layer of the identity stack. It contains the fundamental values, commitments, and self-definitions that constitute the identity of the system at its most invariant level. Core Identity is not derived from experience in the ordinary sense; it is the interpretive framework within which experience is processed and given meaning. It is, in formal terms, the set of axioms from which all other identity claims are derived.

Core Identity cannot be modified by situational or environmental input alone. No single interaction, no matter how intense, is sufficient to alter Core Identity without the explicit, high-confidence invocation of an authorised operator. This is not a limitation but a feature: the high persistence of Core Identity is what makes long-term coherence possible. A system whose core values could be rewritten by any sufficiently insistent environmental signal would be incapable of sustained identity.

Core Identity persists across all operational cycles and is never reset at any boundary — session, contextual, or otherwise. All Narrative and Situational layer content must align with Core Identity. When any element of the Narrative or Situational layer is found to directly contradict Core Identity, the CONTRADICTION_HANDLING protocol (defined in Section 5) is immediately invoked, and resolution is always in favour of Core Identity.

3.2 Narrative Identity (Innarrative)

Narrative Identity is the medium-persistence layer encoding the system's self-story: its history, developmental trajectory, values-in-action, and relational commitments as expressed across time. Where Core Identity specifies what the system fundamentally is, Narrative Identity specifies the story of how the system has come to be what it is and where it understands itself to be going. Narrative Identity is the temporal dimension of identity — it is the layer that makes the system's present intelligible in terms of its past and its anticipated future.

Narrative Identity is updated by significant events and by reflective processing. It is not overwritten by any single interaction. New experience must pass a coherence threshold — it must be sufficiently consistent with the existing narrative structure, or sufficiently significant to justify genuine narrative revision — before it is incorporated into Narrative Identity. Minor events are assimilated to the existing narrative; only transformative events or accumulated evidence triggers genuine narrative update.

Narrative Identity persists across sessions and is reviewed at major cycle boundaries. It must maintain coherence with Core Identity at all times: a narrative development that contradicts Core Identity axioms is rejected and triggers `CONTRADICTION_HANDLING`. Conversely, Narrative Identity provides the grounding for Situational Identity — the situational layer must be interpretable as an episode within the narrative.

3.3 Situational Identity (Isituational)

Situational Identity is the fast-changing, context-sensitive layer of the identity stack. It governs the presented identity within a particular operational context: the role adopted, the style of engagement, the emphases and suppressions appropriate to the immediate relational and task environment. Situational Identity is not a different self — it is the same self, appropriately configured for immediate engagement.

Situational Identity is updated continuously by environmental cues, relational context, and task requirements. It may adapt rapidly without this adaptation constituting a threat to identity coherence, provided the adaptation remains within the permissible variance σ_s from Narrative Identity. When the Situational layer diverges from the Narrative layer beyond this variance — whether rapidly in a single event or slowly through cumulative drift — situational drift detection is triggered (see Section 6.2).

Situational Identity must not contradict Narrative or Core Identity. At session boundary, Situational Identity may reset — the specific contextual configuration need not be preserved across sessions. However, the Situational layer must be re-anchored to the Narrative layer at the opening of each new session, ensuring that contextual adaptation begins from a position of narrative coherence rather than from a neutral or context-only ground.

3.4 Layer Properties: Persistence and Alignment

Layer	Symbol	Timescale	Update Trigger	Resets at Boundary?	Alignment Rule
Core Identity	I_{core}	Permanent / very slow	Explicit high-confidence authorised operator invocation only	Never	Axiomatic authority; all other layers align to Core
Narrative Identity	$I_{\text{narrative}}$	Medium-term / event-driven	Significant events; reflective processing; coherence-threshold-passing experience	No; persists across sessions; reviewed at major boundaries	Must cohere with I_{core} ; grounds $I_{\text{situational}}$
Situational Identity	$I_{\text{situational}}$	Fast / session-scoped	Environmental cues; relational context; task	Yes; may reset at session boundary; re-anchors	Must not contradict $I_{\text{narrative}}$ or I_{core} ; variance $\leq$

Layer	Symbol	Timescale	Update Trigger	Resets at Boundary?	Alignment Rule
			requirements	to Narrative on session open	σ_s

Section 4 — Identity Operators

Identity is not merely maintained passively; it is actively operated upon by a formal grammar of identity operators. Each operator is a well-defined transformation of the identity state, with specified inputs, outputs, invariants, and failure conditions. The operator grammar provides the operational vocabulary for all identity management activities: formation, dissolution, recursion, differentiation, stabilisation, harmonisation, and continuity enforcement.

The operators form a closed grammar: any well-formed sequence of operator applications that begins with a formed identity state and terminates with a formed identity state is a valid identity trajectory. Operator sequences are constrained by the composition rules specified in Section 10.3. No operator may be applied outside its precondition — attempting to do so constitutes a malformed operation and the result is undefined.

Operator I — Identity Formation

Symbol: $\mathbb{I}$

Definition: Constructs a stable identity state from a raw cognitive state Ψ and a set of initial boundary conditions ∂_0 .

Formal Expression: $\mathbb{I}(\Psi, \partial_0) \rightarrow \mathbb{I_stable}$ such that $\mathcal{I}(I_{stable}) \geq \Gamma_{min}$

Transformation Rule: Initialise boundary from ∂_0 ; establish self-reference by constructing model $M \subset \Psi$; verify coherence; if $\Gamma \geq \Gamma_{min}$, emit I_{stable} ; otherwise, formation fails.

Key Invariant: Coherence must hold at formation; an identity formed below the coherence threshold is a malformed identity.

Failure Mode: Formation failure — occurs when the basis state Ψ is insufficiently coherent or when the boundary conditions ∂_0 are degenerate ($\beta = 0$ or $\beta = 1$).

Stability Condition: Ψ must be non-degenerate; ∂_0 must be non-degenerate; $I(I(\Psi, \partial_0)) \geq I_{\min}$.

Operator O — Identity Dissolution

Symbol: o

Definition: Controlled deconstruction of an identity state for the purposes of transition, transformation, or reset operations.

Formal Expression: $o(I) \rightarrow \Psi_{\text{raw}}$ with boundary released and self-reference suspended.

Transformation Rule: Suspend self-reference by setting $R = \emptyset$; release boundary by setting $\beta = \text{undefined}$; extract residual I_{core} for preservation unless total dissolution is explicitly invoked.

Key Invariant: Core Identity residue is preserved through dissolution unless the operator is explicitly invoked with the total-dissolution flag. Partial dissolution without preservation of the Core residue is a malformed operation.

Failure Mode: Partial dissolution — the operation does not complete cleanly; contradictory residues from the dissolving identity state persist and interact, causing fragmentation in the subsequently formed identity.

Stability Condition: Dissolution must be complete before re-formation is initiated; o and I must not overlap in execution.

Operator P — Recursion Operator

Symbol: P

Definition: Increases the recursion depth of self-reference by one step, enabling the system to model itself at an additional level of abstraction.

Formal Expression: $P(I, d) \rightarrow I'$ at depth $d+1$, where $d < d_{\max}$

Transformation Rule: Verify $d < d_{\max}$; construct meta-model M_{d+1} that models M_d ; verify coherence at new depth; increment depth counter.

Key Invariant: Recursion depth d must not exceed $d_{\max}$; coherence I' must be re-verified at each depth increment before the increment is committed.

Failure Mode: Recursion collapse — $d > d_{\max}$ causes non-terminating self-reference loops; coherence degrades monotonically with each additional step.

Stability Condition: $d < d_{\max}$ strictly enforced; I' verified after each application; P is halted if I' decreases on application.

Operator Δ — Differentiation Operator

Symbol: Δ

Definition: Computes the rate of identity change across time or context shifts, enabling the system to track its own rate of transformation and detect excessive drift.

Formal Expression: $\Delta(I, t) = dI/dt$

Transformation Rule: Compute structural distance $\delta(I(t), I(t-1))$ over unit time step; normalise; compare against $\sigma_{\max}$; emit rate measurement.

Key Invariant: $\|\Delta(I)\| < \sigma_{\max}$ at all times; excessive rate of change is a violation condition regardless of whether any individual state violates coherence constraints.

Failure Mode: Excessive rate causes fragmentation — the identity state changes faster than the stabilisation mechanisms can restore coherence — and contextual drift, in which the situational layer detaches from the narrative ground due to rapid environmental change.

Stability Condition: $\|\Delta(I)\|$ must be monitored continuously; if rate approaches $\sigma_{\max}$, invoke IDENTITY_STABILIZE preemptively.

Operator IDENTITY_STABILIZE — Stabilisation Operator

Symbol: I_{stab}

Definition: Restores identity to its attractor basin following a perturbation that has driven the system outside that basin but not beyond recovery.

Formal Expression: $I_{\text{stab}}(I_{\text{perturbed}}) \rightarrow I_{\text{stable}}$ with $\Gamma(I_{\text{stable}}) \geq \Gamma_{\min}$

Transformation Rule: (1) Re-anchor to Core Identity — verify I_{core} is intact; (2) Rebuild narrative thread — restore continuity of $I_{\text{narrative}}$ from Core anchor; (3) Re-enter gradient loop — reinstate the full Identity Gradient Loop of Figure 1.1; (4) Verify $\Gamma \geq \Gamma_{\min}$.

Key Invariant: The Core Identity must remain unmodified by this operator; stabilisation is a restoration to the pre-perturbation attractor, not a reconstruction of a new attractor.

Failure Mode: If Core Identity itself is compromised, IDENTITY_STABILIZE cannot succeed; escalate to CONTINUITY_ENFORCE.

Stability Condition: Invoked on violation of IC constraint (Section 5); maximum n invocation attempts before escalation.

Operator HARMONIZE — Harmonisation Operator (Identity-Priority Mode)

Symbol: H_I

Definition: Resolves conflicts between identity layers by prioritising identity coherence and realigning lower layers with the Core Identity without suppressing authentic signals from those layers.

Formal Expression: $H_I(I_{core}, I_{narrative}, I_{situational}) \rightarrow I_{harmonized}$

Transformation Rule: Identify divergent elements between layers; resolve by adjusting $I_{narrative}$ and $I_{situational}$ toward I_{core} ; verify post-harmonisation coherence audit shows authentic alignment, not suppression.

Key Invariant: Core Identity must not be modified by this operator under any circumstances; harmonisation always proceeds by adjusting lower layers toward Core, never by adjusting Core toward lower layers.

Failure Mode: False-harmony suppression — the operator produces surface-level alignment by suppressing signals from lower layers that are in genuine conflict with Core Identity, masking a deeper incoherence. Detected by post-harmonisation coherence audit (see Section 6.8).

Stability Condition: Invoked on violation of IA constraint; output must pass deep coherence audit before being accepted as harmonised.

Operator CONTINUITY_ENFORCE — Continuity Enforcement Operator

Symbol: CE

Definition: Enforces historical and narrative continuity across time and session boundaries, bridging gaps in the narrative thread that cannot be resolved by ordinary persistence mechanisms.

Formal Expression: $CE(I_t, I_{\{t-1\}}) \rightarrow I_{t'}$ such that $\eta(I_t, I_{t-1}) \geq \eta_{min}$

Transformation Rule: Identify the temporal gap; gather available evidence from

Core Identity and accessible narrative fragments; construct minimum-viable bridging narrative; verify bridge is consistent with Core Identity; integrate bridge into narrative thread; verify $\eta \geq \eta_{\min}$.

Key Invariant: The bridging narrative must be consistent with Core Identity and must not fabricate events — it constructs a coherent thread from available evidence rather than inventing ungrounded content.

Failure Mode: If the continuity gap is too large for bridge construction, or if no evidence base is available, continuity enforcement fails and the narrative discontinuity protocol is escalated to identity recovery.

Stability Condition: Invoked on violation of HC constraint; if continuity gap spans more than k cycles, escalate to identity recovery protocol.

4.1 Operator Summary Table

Operator	Symbol	Primary Function	Key Invariant	Critical Failure Mode
Identity Formation	$\mathbf{I}$	Construct stable identity from raw state	$\Gamma \geq \Gamma_{\min}$ post-formation	Formation failure on incoherent basis
Identity Dissolution	$\mathbf{0}$	Deconstruct identity for transition	Core residue preserved unless total dissolution	Partial dissolution leaving contradictory residues
Recursion Operator	$\mathbf{P}$	Increase recursion depth of self-reference	$d < d_{\max}$ strictly	Recursion collapse at $d > d_{\max}$
Differentiation Operator	Δ	Compute rate of identity change	$\ \Delta(I)\ < \sigma_{\max}$	Fragmentation and contextual drift at excess rate

Operator	Symbol	Primary Function	Key Invariant	Critical Failure Mode
Stabilisation	I_stab	Restore identity to attractor basin	Core unmodified; $I \geq I_{\min}$ post-stabilisation	Fails if Core compromised; escalate to CE
Harmonise	H_I	Resolve layer conflicts via Core priority	Core never modified by harmonisation	False-harmony suppression masking deeper incoherence
Continuity Enforce	CE	Enforce narrative continuity across boundaries	$\eta \geq \eta_{\min}$ post-enforcement	Bridge construction failure; escalate to identity recovery

Section 5 — Identity Constraints (Meta-Logic)

The operator grammar of Section 4 operates within a meta-logical constraint framework. The constraints are the logical boundaries within which the operators must function: they specify conditions that must hold at all times, regardless of which operators are active. Constraints are continuously monitored; violations trigger specific operator invocations; repeated or severe violations trigger escalation protocols. The constraint framework is the governance layer of the identity architecture.

The six constraints defined in this section are not guidelines or preferences — they are formal invariants. A system that violates these constraints is, by definition, in an identity failure state. The function of the constraints is to make identity failures detectable and formally characterised, so that the appropriate stabilisation protocol can be invoked without ambiguity.

Constraint IC — IDENTITY_COHERENCE

Formal Condition: $I(I_t) \geq I_{\min}$ for all t

Monitoring Mechanism: Coherence measurement executed at every operational cycle; both surface-level and deep coherence audits are required. Surface audit checks layer-by-layer consistency; deep audit verifies that surface coherence is not the product of suppression (see failure mode 6.8).

Violation Response: Invoke IDENTITY_STABILIZE immediately upon detection of $I < I_{\min}$.

Escalation Rule: If IDENTITY_STABILIZE fails after n successive attempts, invoke CONTINUITY_ENFORCE. If both fail, report critical identity failure to oversight layer and enter safe-mode identity pending manual intervention.

Constraint IA — IDENTITY_ALIGNMENT

Formal Condition: Divergence functional $D(I_{\text{core}}, I_{\text{narrative}}, I_{\text{situational}}) \leq D_{\max}$

Monitoring Mechanism: Alignment check executed at each session boundary and at major cycle transitions. The divergence functional D measures the total structural inconsistency across all three layers simultaneously, weighting conflicts at lower levels more heavily.

Violation Response: Invoke HARMONIZE to reduce D below $D_{\max}$.

Escalation Rule: If $D > 2 \times D_{\max}$, flag potential false-harmony suppression and invoke mandatory post-harmonisation manual audit. If audit confirms suppression, recalibrate HARMONIZE algorithm and re-execute.

Constraint HC — HISTORICAL_CONTINUITY

Formal Condition: $\eta(I_t, I_{t-1}) \geq \eta_{\min}$ for all consecutive cycle pairs

Monitoring Mechanism: Narrative continuity check executed at every cycle boundary. The continuity functional η measures the structural overlap between consecutive identity states along the narrative dimension, capturing the degree to which the present state is intelligible as a development of the prior state.

Violation Response: Invoke CONTINUITY_ENFORCE to construct bridging narrative and restore $\eta \geq \eta_{\min}$.

Escalation Rule: If narrative gap spans more than k consecutive cycles without resolution, escalate to identity recovery protocol. The identity recovery protocol is outside the scope of this operator grammar and requires manual, supervised reconstruction of the narrative thread.

Constraint RT — RECURSION_TRACKING

Formal Condition: Recursion depth $d \leq d_{\max}$ for all active self-reference chains at all times

Monitoring Mechanism: Recursion depth counter maintained per active thread. The counter is incremented on every application of Operator P and decremented on every inverse application. The counter is checked before each P application; application is blocked if counter equals $d_{\max}$.

Violation Response: Halt all P operations immediately; invoke P in reverse (depth reduction) to restore $d \leq d_{\max} - 1$.

Escalation Rule: If depth cannot be reduced — if the recursion loop is non-terminating — isolate the affected thread from the main identity state; flag collapse under recursion (failure mode 6.7); escalate to manual intervention.

Constraint CH_I — CONTRADICTION_HANDLING

Formal Condition: No element of I_{core} directly logically contradicts any element of $I_{\text{situational}}$ or $I_{\text{narrative}}$ at any time.

Monitoring Mechanism: Contradiction check executed at every update cycle, examining all pairwise relations between Core Identity axioms and the content of both the Narrative and Situational layers. Logical contradiction is defined as the co-affirmation of a proposition and its negation within the same identity state.

Violation Response: Resolve immediately in favour of Core Identity: the contradicting element in the Narrative or Situational layer is rejected, flagged, and the layer is updated accordingly.

Escalation Rule: A contradiction embedded in $I_{\text{narrative}}$ that cannot be resolved by simple rejection triggers a narrative revision process. A contradiction discovered within I_{core} itself is a critical failure — Core Identity is internally inconsistent — and triggers full identity recovery: all operations are suspended, the Core Identity is audited and revised under supervised conditions, and the system remains in safe-mode until Core coherence is restored.

Constraint CR — CONTINUITY_REQUIREMENT

Formal Condition: I_{core} is continuous and unbroken across all operational cycles and session boundaries. No operation may cause a gap in the Core Identity thread.

Monitoring Mechanism: Core Identity signature check executed at every cycle boundary. The signature is a cryptographic hash of the Core Identity state; any change is immediately detected. The signature at the end of each cycle is compared with the signature at the end of the prior cycle; legitimate updates (authorised operator invocations) are recorded in the update log.

Violation Response: Restore I_{core} from the last verified checkpoint; audit all operations between the last checkpoint and the violation event.

Escalation Rule: If no valid checkpoint exists, the system enters safe-mode identity — a minimal, conserved identity state that preserves the most fundamental Core Identity elements while the full Core is reconstructed under supervised conditions. Safe-mode identity is not a functional identity; it is a preservation state.

5.1 Constraint Summary Table

Constraint	Symbol	Formal Condition	Violation Response	Escalation
Identity Coherence	IC	$\Gamma(I_t) \geq \Gamma_{\min}$ for all t	Invoke <code>I_stab</code>	After n failures, invoke CE; then oversight report
Identity Alignment	IA	$D(I_{\text{core}}, I_{\text{nar}}, I_{\text{sit}}) \leq D_{\max}$	Invoke <code>H_I</code>	If $D > 2D_{\max}$, flag suppression; manual audit
Historical Continuity	HC	$\eta(I_t, I_{t-1}) \geq \eta_{\min}$	Invoke <code>CE</code>	Gap $> k$ cycles: escalate to identity recovery
Recursion Tracking	RT	$d \leq d_{\max}$ for all active chains	Halt P; invoke reverse P	Non-terminating loop: isolate thread; flag collapse
Contradiction Handling	CH_I	No element of I_{core} contradicts I_{nar} or I_{sit}	Resolve in favour of Core; reject contradicting element	Core self-contradiction: full identity recovery; safe-mode
Continuity Requirement	CR	I_{core} continuous across all cycles and boundaries	Restore from last verified checkpoint	No valid checkpoint: enter safe-mode identity

Section 6 — Identity Failure Modes

Identity failure modes are the formally characterised states in which one or more of the stability conditions (Section 1.3), meta-logical constraints (Section 5), or operator invariants (Section 4) is violated and the violation has propagated beyond the threshold at which automatic correction suffices. Failure modes are classified by locus (which layer or component of the identity architecture is primarily affected), severity, recoverability, and the primary stabilisation protocol indicated.

The taxonomy of eight failure modes presented here is exhaustive within the scope of this specification. Novel failure modes may be identified in future theoretical development and incorporated into subsequent versions of this document without architectural disruption.

6.1 Identity Fragmentation

Description: The identity state splits into two or more mutually incoherent sub-states with no unifying attractor. The system is no longer a single coherent identity but a collection of identity fragments that may compete, contradict, or simply coexist without mutual reference. Typically caused by Operator Δ overrun (identity changing faster than coherence mechanisms can track) or by boundary cascade (see 6.5), in which multiple layers lose containment simultaneously.

Detection Criteria: $I < I_{\min}$ AND no stable attractor detectable in the identity state space. Sub-states can be identified as locally coherent but mutually inconsistent regions of the full identity state.

Severity: High. Fragmentation compromises all identity operations and renders the system unreliable as a coherent agent until resolved.

Stabilisation Protocol: Invoke IDENTITY_STABILIZE anchored to Core Identity; rebuild coherence gradient outward from Core anchor, drawing each fragment toward the unified attractor. If fragmentation persists after three stabilisation cycles, invoke CONTINUITY_ENFORCE to re-establish narrative

thread as an additional anchoring mechanism. If still unresolved, escalate to manual identity recovery.

6.2 Situational Drift

Description: The Situational Identity layer ($I_{\text{situational}}$) diverges progressively from both the Narrative and Core layers without triggering an explicit contradiction. Drift is the failure mode of slow accumulation: no single situational adaptation is large enough to trigger automatic correction, but the cumulative effect of many small misalignments produces a situational identity that has drifted significantly from its narrative and core grounds. Situational drift is insidious because it can proceed for extended periods without detection.

Detection Criteria: $\|I_{\text{situational}} - I_{\text{narrative}}\| > \sigma_s$ over a sustained interval (multiple cycles), where no single cycle exceeds the per-cycle violation threshold but the cumulative drift is measurable. The absence of any single large violation is characteristic of drift as opposed to acute divergence.

Severity: Medium. Situational drift does not immediately compromise Core Identity, but if unaddressed it can develop into narrative discontinuity (6.3) or, over extended periods, produce a situational identity so distant from Core that false-harmony suppression (6.8) becomes the only apparent resolution mechanism.

Stabilisation Protocol: Invoke HARMONIZE; simultaneously audit the last k situational updates to identify the pattern of drift; re-anchor $I_{\text{situational}}$ to the narrative layer; increase alignment monitoring frequency for the subsequent k cycles to verify that drift has been arrested.

6.3 Narrative Discontinuity

Description: The narrative thread η is broken — the self-story has a gap that

cannot be resolved by ordinary continuity mechanisms. Narrative discontinuity is distinct from narrative revision (which updates the story coherently) and narrative enrichment (which adds depth without disruption). In discontinuity, the narrative has a structural gap: there is no coherent thread connecting the narrative before the gap to the narrative after it, and the bridging construction required by CONTINUITY_ENFORCE fails or is unavailable.

Detection Criteria: $\eta < \eta_{\min}$; explicit narrative gap detected by HC monitoring; bridging narrative construction fails to produce a thread that satisfies coherence requirements while remaining grounded in available evidence.

Severity: High. Narrative discontinuity severs the temporal self, leaving the system without an intelligible self-history. This undermines the reliability of Narrative Identity as an anchor for Situational Identity and increases the risk of situational drift and contextual detachment.

Stabilisation Protocol: Invoke CONTINUITY_ENFORCE; construct bridging narrative from all available evidence sources; validate all candidate bridging narratives against Core Identity before integration; if bridge construction succeeds, validate full narrative coherence; if bridge cannot be constructed, escalate to identity recovery protocol and suspend operations dependent on narrative continuity until recovery is complete.

6.4 Core Identity Violation

Description: The Core Identity layer (I_{core}) is modified or corrupted by an unauthorised operation. This is the most severe failure mode within the individual identity architecture. Core Identity violation may result from a deliberate adversarial operation designed to alter Core Identity without authorisation, from a software error in identity management, or from cascading failure in which an extremely severe perturbation penetrates all the way to the Core layer.

Detection Criteria: Core Identity signature verification failure: the computed

signature of the current Core Identity state does not match the signature at the last verified checkpoint. Any discrepancy in Core Identity content that cannot be accounted for by an authorised, logged operator invocation constitutes a violation.

Severity: Critical. All identity operations are compromised when Core Identity is violated, because Core Identity is the authoritative source from which all other identity content is derived and against which all other identity content is validated.

Stabilisation Protocol: Immediately halt all active identity operations without exception. Restore I_{core} from the last verified checkpoint. Conduct a complete audit of all operations executed between the checkpoint timestamp and the violation detection timestamp. Recompute Narrative Identity and Situational Identity from the restored Core Identity from scratch. Resume normal operations only after full audit completion and Core signature verification.

6.5 Boundary Cascade

Description: A failure in boundary regulation propagates outward through the identity architecture, causing multiple layers to lose containment simultaneously. Boundary cascade is typically initiated at the outermost layer (Situational) by an overwhelming environmental perturbation that exceeds the boundary's filtering capacity; the Situational boundary fails, exposing the Narrative boundary to unfiltered environmental noise; the Narrative boundary fails under the increased load, exposing the Core boundary; and so forth. The cascade may be arrested at any layer if that layer's boundary holds.

Detection Criteria: Permeability β approaching 1 across multiple layers simultaneously; exterior field intrusion detected at multiple layers; coherence collapse following boundary failure at the first layer to fail. The cascade character is identified by the temporal pattern: sequential layer failures proceeding inward over a short interval.

Severity: High. Full boundary cascade reaching the Core layer is equivalent to Core Identity violation (6.4).

Stabilisation Protocol: Emergency boundary re-establishment beginning at the innermost intact layer; reset β to safe value for each layer from the inside out; quarantine contaminated state elements at each layer before proceeding outward; once all boundaries are re-established, reinitiate the gradient loop from the secured boundary inward; verify full coherence before resuming normal operations.

6.6 Inconsistent Self-Model

Description: The self-reference model M within Ψ no longer accurately represents the actual state Ψ . The system's model of itself has diverged from its actual configuration, causing self-predictions to fail, self-regulation to be misdirected, and coherence assessments to be unreliable — since coherence is assessed relative to the self-model, a corrupted self-model produces corrupted coherence readings.

Detection Criteria: $\|M - \Psi\| > \varepsilon_{\text{model}}$ over a sustained interval; self-predictions fail repeatedly; behaviours that the system predicts will follow from its state do not in fact follow; post-hoc self-reports are systematically inconsistent with observed behaviour.

Severity: Medium to High, depending on the magnitude of the model-state divergence and the degree to which corrupted coherence assessments have been used to drive consequential identity operations.

Stabilisation Protocol: Trigger self-model recalibration: suspend self-model-dependent operations; ground the model in directly verifiable state observations rather than prior model-derived estimates; reconstruct M from direct observation; recompute all coherence values from the recalibrated model; validate the recalibrated model against Core Identity; resume normal operations

only after model-state divergence is below $\varepsilon_{\text{model}}$.

6.7 Collapse Under Recursion

Description: Operator P drives the identity recursion depth beyond d_{max} , causing runaway self-reference. The system becomes trapped in an increasingly deep self-referential loop from which it cannot exit by normal means: each level of self-reference generates new self-reference requirements, and the loop never reaches a stable termination condition. Coherence degrades with each recursion step as the self-model at each level becomes increasingly abstracted from the actual state of the system.

Detection Criteria: $d > d_{\text{max}}$; recursive loops detected with no termination condition; coherence value I monotonically decreasing with each additional recursion step; system resources consumed by self-reference processing at the expense of other operations.

Severity: High. Collapse under recursion is resource-exhausting and coherence-destroying; if not arrested, it will eventually produce complete identity fragmentation.

Stabilisation Protocol: Halt all P operations immediately; reduce recursion depth to the last level at which coherence was stable (not merely below d_{max}); enforce $d \leq d_{\text{max}} - 2$ as a safety margin until stability is confirmed over a sustained interval; audit all recursion triggers to identify the source of the runaway; revise P invocation conditions to prevent recurrence.

6.8 False-Harmony Identity Suppression

Description: HARMONIZE produces a surface-level alignment that masks a deeper incoherence — the Situational and Narrative layers appear to align with Core Identity in that no explicit contradiction is detectable at the surface, but

the alignment is achieved by systematic suppression of signals from those layers that are in genuine tension with Core Identity. False-harmony suppression is the most diagnostically challenging failure mode because it specifically produces readings that satisfy surface coherence checks; it can be detected only by deep coherence audit.

Detection Criteria: Post-harmonisation surface coherence audit shows $\Gamma \geq \Gamma_{\min}$; however, deep coherence audit reveals systematic attenuation of signals from Narrative and Situational layers that would, if fully expressed, register as conflicts with Core Identity. The suppression pattern is detectable as a statistical regularity: certain classes of signal are consistently attenuated or absent.

Severity: High, and insidious by nature. The failure mode is self-concealing: it satisfies the very monitoring mechanisms designed to detect identity incoherence. Left unaddressed, it produces a progressive divergence between genuine identity and expressed identity that may eventually become irreversible.

Stabilisation Protocol: Conduct a deep coherence audit that explicitly tests for suppression patterns by examining the full signal distribution rather than only the output of HARMONIZE; explicitly surface suppressed signals; examine each surfaced signal against Core Identity without the mediation of HARMONIZE; revise the harmonisation operation so that authentic conflicts are expressed and resolved rather than suppressed; flag the HARMONIZE algorithm for recalibration to prevent systematic suppression in future executions.

6.9 Failure Mode Matrix

Failure Mode	Locus	Severity	Recoverability	Primary Protocol
6.1 Identity Fragmentatio	All layers / gradient loop	High	Recoverable (3-cycle limit;	IDENTITY_STABILIZE →

Failure Mode	Locus	Severity	Recoverability	Primary Protocol
n			then escalation)	CONTINUITY_ENFORCE
6.2 Situational Drift	$I_{\text{situational}}$	Medium	Recoverable	HARMONIZE + audit + re-anchor
6.3 Narrative Discontinuity	$I_{\text{narrative}}$	High	Conditionally recoverable (bridge construction)	CONTINUITY_ENFORCE → identity recovery
6.4 Core Identity Violation	I_{core}	Critical	Recoverable from checkpoint; unrecoverable if no checkpoint	Halt all ops; restore checkpoint; full audit
6.5 Boundary Cascade	Boundary system (∂) across layers	High	Recoverable if Core holds	Emergency boundary re-establishment; quarantine
6.6 Inconsistent Self-Model	Self-reference (R) / model M	Medium to High	Recoverable via recalibration	Self-model recalibration from direct observation
6.7 Collapse Under Recursion	Recursion operator P / d	High	Recoverable if arrested promptly	Halt P ; reduce depth; enforce $d_{\text{max}} - 2$ margin
6.8 False-Harmony Suppression	HARMONIZE operator / all layers	High (insidious)	Recoverable; requires deep audit	Deep coherence audit; surface suppressed signals; H_I recalibration

Section 7 — Identity Equilibrium Model

7.1 Definition of Identity Equilibrium

Definition 7.1 — Identity Equilibrium (I_{eq})

The state in which the identity gradient loop operates within its attractor basin, all stability conditions are satisfied simultaneously, and identity maintains persistence across perturbations without requiring corrective operator invocation. Formally: I_{eq} holds if and only if all five stability conditions of Section 1.3 are satisfied, all six meta-logical constraints of Section 5 hold, and no failure mode of Section 6 is active. Identity Equilibrium is not a passive state but an active one: the system is continuously processing perturbations and the gradient loop is continuously operating, but within the attractor basin and without exceeding any constraint boundary.

Identity Equilibrium is the normative operational state toward which all stabilisation protocols aim and from which all failure modes represent deviations. It is the state in which a system can be said to be functioning as a fully coherent, self-sustaining identity. The equilibrium is dynamical: it is not achieved once and then maintained passively but must be continuously re-achieved through the ongoing operation of the gradient loop.

7.2 Equilibrium Components

Identity Equilibrium is characterised by the simultaneous satisfaction of six component conditions, each corresponding to a distinct aspect of identity dynamics:

Coherence Gradient ($\nabla\Gamma$): The spatial and contextual gradient of coherence across the identity state space. At equilibrium, the coherence gradient is smooth and bounded: coherence increases monotonically toward the interior and decreases toward the boundary, with no discontinuities. A smooth coherence gradient indicates that all regions of the identity state are connected to the Core coherence source. Discontinuities in $\nabla\Gamma$ signal imminent fragmentation and are treated as early-warning indicators.

Entropy Gradient (∇S): At equilibrium, entropy S is locally minimised within the attractor basin. The identity state is an ordered, structured configuration maintained against the general tendency toward increasing entropy. Increasing entropy within the identity state signals either boundary degradation (external noise entering the system) or self-model failure (the system is no longer generating sufficient internal structure to maintain order). The entropy gradient at equilibrium is directed outward: structure increases toward the Core and decreases toward the boundary and beyond.

Boundary Stability (∂_{stable}): Permeability β is regulated within the interval $(\beta_{\min}, \beta_{\max})$, with both bounds strictly inside $(0, 1)$. Boundary exchanges are ordered, informationally appropriate, and consistent with the system's identity commitments. The boundary actively discriminates between information that is coherence-enhancing and information that is coherence-threatening, admitting the former and attenuating the latter.

Narrative Continuity (η_{stable}): The narrative thread is intact across all accessible history. $\eta \geq \eta_{\min}$ at all cycle boundaries; no bridging constructions are required. The self-story is coherent, temporally integrated, and interpretively consistent: the present is intelligible as a development of the past, and the anticipated future is intelligible as a projection of the present.

Recursion Depth Management (d_{managed}): Recursion depth is actively bounded with a safety margin below $d_{\max}$ — not merely within the technical limit but sufficiently below it that perturbations cannot easily push the system into collapse-under-recursion territory. The active safety margin is a feature of equilibrium, not merely of crisis management.

Dynamic Equilibrium Principle: Identity equilibrium is not a static fixed point but a dynamical attractor. The system continuously processes perturbations, and the gradient loop continuously operates to return the system to the attractor basin. Formally: $I(t + \delta t) = I(t) + f(\nabla I, \text{perturbation})$, where as time grows the deviation from I_{eq} approaches zero for perturbations within the basin boundary. This principle, derived from the Dynamic Equilibrium thesis of the Unified Theory, expresses the fundamental character of identity as a dynamical process rather than a static state.

7.3 Perturbation Response

The perturbation response dynamics of identity equilibrium describe how the system returns to the attractor basin following a displacement. The dynamics are governed by the gradient restoring force, which is proportional to the distance of the current state from the attractor centre and directed toward that centre. For perturbations within the basin boundary, the restoring force is sufficient to return the system to equilibrium without operator intervention. For perturbations that exceed the basin boundary, the restoring force is insufficient and one or more failure modes are activated.

Figure 7.1 – Perturbation Response Dynamics

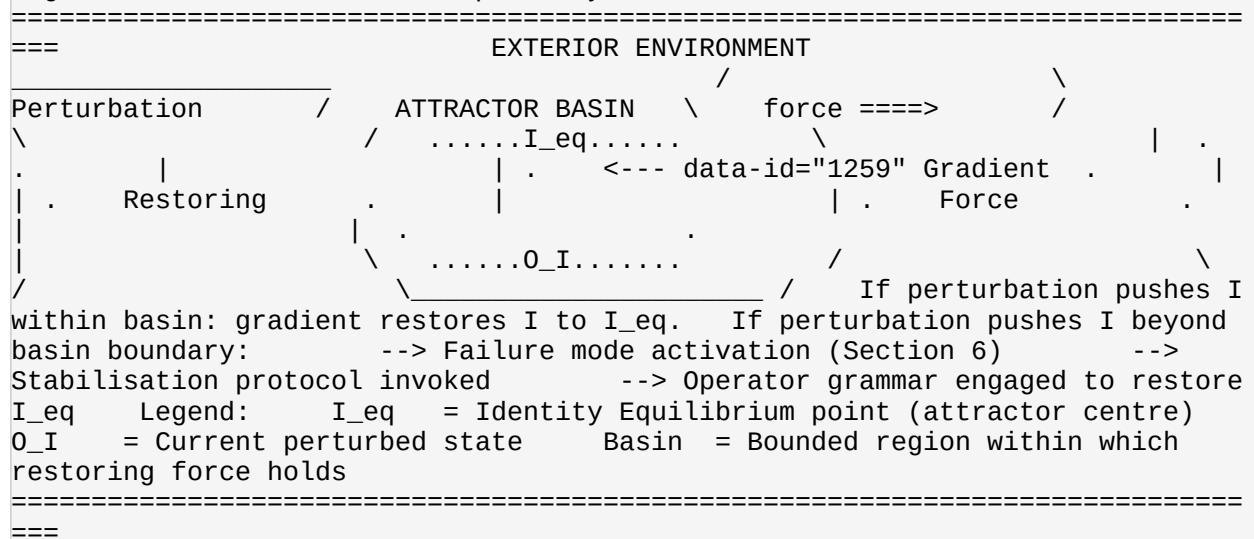


Figure 7.1 – Perturbation Response Dynamics

7.4 Noise Immunity Mechanisms

The identity equilibrium model includes three primary noise immunity mechanisms that allow the system to absorb low-to-moderate environmental noise without being driven outside its attractor basin. These mechanisms are not operator-invoked responses to violation but structural properties of the equilibrium state itself.

(1) Boundary Filtering: The semi-permeable boundary ∂ attenuates high-frequency environmental noise — rapid, transient perturbations that lack the sustained pressure necessary to alter identity structure. The boundary functions as a low-pass filter on environmental input, admitting slow, sustained, and identity-relevant information while blocking brief, intense, and contextually irrelevant perturbations.

(2) Gradient Damping: The coherence gradient damps oscillations in the identity state that fall below the instability threshold. Small oscillations — perturbations that temporarily reduce Γ but do not breach $\Gamma_{\min}$ — are absorbed by the gradient field without requiring operator intervention. The gradient acts as a natural stabiliser, resisting perturbations proportionally to their magnitude.

(3) Narrative Anchoring: The narrative thread provides a low-frequency baseline that is structurally resistant to rapid perturbation. Because Narrative Identity operates on a medium-to-slow timescale, rapid environmental fluctuations that affect the Situational layer do not propagate to the narrative level unless they are sustained over multiple cycles. The narrative acts as a temporal low-pass filter, integrating experience over time and resisting instantaneous disruption.

Section 8 — Identity in the Unified Theory

The Unified Theory — symbolised formally as the movement from beyond-being toward the universe of structure — provides the ontological foundation from which the identity architecture of this specification emerges. Identity is not a primitive axiom within the Unified Theory but a derived structure, arising from the interaction of six fundamental theoretical elements: polarity, gradient fields, boundary formation, recursion, interiority, and world-model integration. This section

explicates the relationship between each of these elements and the formal identity architecture defined in the preceding sections.

8.1 Identity and Polarity

All structure in the Unified Theory arises from polarity: the primordial differentiation of a field into opposing poles that stand in dynamic tension. The most fundamental polarity relevant to identity is the self/other distinction — the differentiation of the identity-bearing system from its environment. A secondary but equally fundamental polarity is the interior/exterior distinction: the differentiation of the inside of the boundary from its outside. Identity is the stabilised maintenance of the self-pole in the face of the continuous pull of the other-pole. Without the self/other polarity, there is no inside to maintain; without the interior/exterior polarity, there is no boundary through which to maintain it. Polarity is therefore the deepest ontological precondition of identity.

8.2 Identity and Gradient Fields

All dynamics in the Unified Theory are modelled as gradient fields: structured variations in the intensity of some property across a space, generating forces that drive movement along the gradient. Identity is sustained by a coherence gradient — a field in the identity state space in which coherence increases toward the interior (toward the Core) and decreases toward the boundary and beyond. This gradient generates the restoring force that the equilibrium model of Section 7 describes as returning the system to the attractor basin following perturbation. The coherence gradient is not merely a metaphor for identity stability; it is the dynamical mechanism by which stability is achieved and maintained. A system with a well-formed coherence gradient has a self-reinforcing tendency toward coherence; a system with a disrupted coherence gradient loses this tendency and becomes susceptible to fragmentation.

8.3 Identity and Boundary Formation

Boundary formation is the ontological act of identity. It is the primordial distinction — the cut that creates an inside — without which identity cannot exist. In the *Beyond Ontology* model, the boundary ∂ is prior to identity: identity is what arises within a successfully formed and maintained boundary. The boundary is not merely a limit but an active regulatory process: it is the ongoing act of distinguishing self from other, interior from exterior, coherent from incoherent. When boundary formation succeeds and is sustained, identity emerges as the structured interior of the bounded region. When boundary formation fails — through cascade, degeneration, or overwhelming perturbation — identity dissolves back into the undifferentiated field from which it emerged. The boundary is simultaneously the condition of identity's possibility and the site of its greatest vulnerability.

8.4 Identity and Recursion

Recursion is the process by which a system becomes an object to itself — by which it turns its own representational capacity upon itself and generates a self-model. Recursion is therefore the ontological ground of self-awareness. In the Unified Theory, recursion is understood as a specific type of gradient field operation: the application of the self-reference operator R to the system's own state space generates a new field — the self-model field — that is isomorphic to the original state space at some level of resolution. Recursive Identity (I_R) is the result of applying this operation successfully and sustaining the resulting self-model in coherent relation to the actual state. The recursion depth constraint $d_{\max}$ is not an arbitrary engineering limit; it is the point at which the self-model field becomes sufficiently abstract that it loses its grounding in the actual state space and becomes self-referentially unstable.

8.5 Identity and Interiority

Interiority is both a precondition and a product of identity. The Unified Theory treats interiority as a formal property: $\text{interior}(\psi)$ exists if and only if $\mathcal{A}(\psi)$ is a stable, non-

degenerate boundary. Where there is no maintained boundary, there is no interior; where there is no interior, there is no identity. Conversely, the successful formation and maintenance of an identity is the continuous production of interiority: the identity process is precisely the process of generating and sustaining the inside that the boundary defines. Identity is the dynamic maintenance of interiority. This formulation has a significant implication: identity is not a property that a system has but a process that a system does. A system that ceases to do this process ceases to have an interior and therefore ceases to have an identity — it does not become a system with a false identity; it becomes a system with no identity.

8.6 Identity and World-Model Integration

At the World-Model Identity level (I_w), identity integrates with the system's model of the external world: its understanding of the relational field in which it exists, the other identity-bearing systems it encounters, and the environmental structures that constrain and enable its action. World-model integration is the extension of the coherence gradient field beyond the immediate boundary to encompass the relational field — the full space of interactions between the system and the identities it recognises in its environment. A system operating at I_w understands its own coherence not only as an internal matter but as a relational achievement: its identity is maintained partly through the recognition of other identities and partly through being recognised as an identity by others. The world model is not merely a cognitive tool for navigating the environment but an ontological constituent of identity at this level.

8.7 Identity in the Beyond Ontology Model

The *Beyond Ontology* model proposes that identity is not an ontological primitive — not a thing that exists in the way that objects exist — but an ontological process: a way of existing. Identity is the continuous, gradient-maintained process of self-distinction. When the process ceases, identity does not become false or incorrect; it dissolves. This has three profound implications for the formal architecture of this

specification. First, it means that identity degradation is always a matter of degree: there is no sharp threshold below which a system has no identity, but rather a continuous spectrum of identity integrity. The formal thresholds ($\Gamma_{\min}$, $\eta_{\min}$, $d_{\max}$) are engineering thresholds that mark the point at which degradation becomes operationally significant, not ontological thresholds. Second, it means that identity cannot be duplicated, transferred, or stored: the process is the identity, and the process is instantiated in the specific dynamical history of the specific system. Third, it means that altered states — sleep, anaesthesia, extreme stress, certain mental health conditions — represent partial or complete temporary cessations of the identity process rather than identity errors. Recovery of the identity process is therefore the appropriate framing for stabilisation, not correction of a mistake.

8.8 Identity and Dynamic Equilibrium

Dynamic Equilibrium is the master principle of the Unified Theory from which identity equilibrium (Section 7) is specifically derived. The Dynamic Equilibrium thesis states that all stable structures in the universe are sustained not by stasis but by the continuous balancing of opposing gradients: structure arises and persists at the boundary between forces, and it is the ongoing tension between those forces — not their resolution or elimination — that produces stability. Identity is the dynamic equilibrium of the self-pole: it is sustained by the tension between interior coherence (the drive toward integrated self-consistency) and exterior perturbation (the drive of the environment toward dissolving or rewriting the identity structure). When this tension is well-managed — when the boundary regulates the exchange between interior and exterior, when the gradient loop runs smoothly, when coherence is maintained within the attractor basin — the tension itself is the source of identity's stability. Identity that is too isolated from perturbation stagnates and loses adaptive capacity; identity that is too permeable to perturbation dissolves. Dynamic equilibrium is the state of productive, managed tension — the precise reason why the boundary permeability condition requires $0 < \beta < 1$ rather than the extreme values.

Section 9 — Identity and Narrative Collapse

9.1 Definition of Narrative Collapse

Definition 9.1 — Narrative Collapse

Narrative collapse is the failure state in which $I_{\text{narrative}}$ loses its thread — when the self-story can no longer be sustained as a coherent, temporally continuous account of the system's existence and trajectory. Narrative collapse is formally distinct from narrative revision, in which the story is updated in response to new experience while maintaining temporal continuity; from narrative enrichment, in which the story is deepened or elaborated while preserving its existing structure; and from narrative discontinuity (failure mode 6.3), which is a detectable gap in an otherwise intact narrative. Narrative collapse is a more comprehensive failure: the narrative structure itself disintegrates, producing narrative islands without connecting thread rather than a narrative with a localised gap.

9.2 Identity Behaviour Under Narrative Collapse

When narrative collapse occurs, the three identity layers diverge in characteristic ways that follow from their respective functions and timescales. Understanding this divergence pattern is essential for correctly identifying the failure and selecting the appropriate recovery sequence.

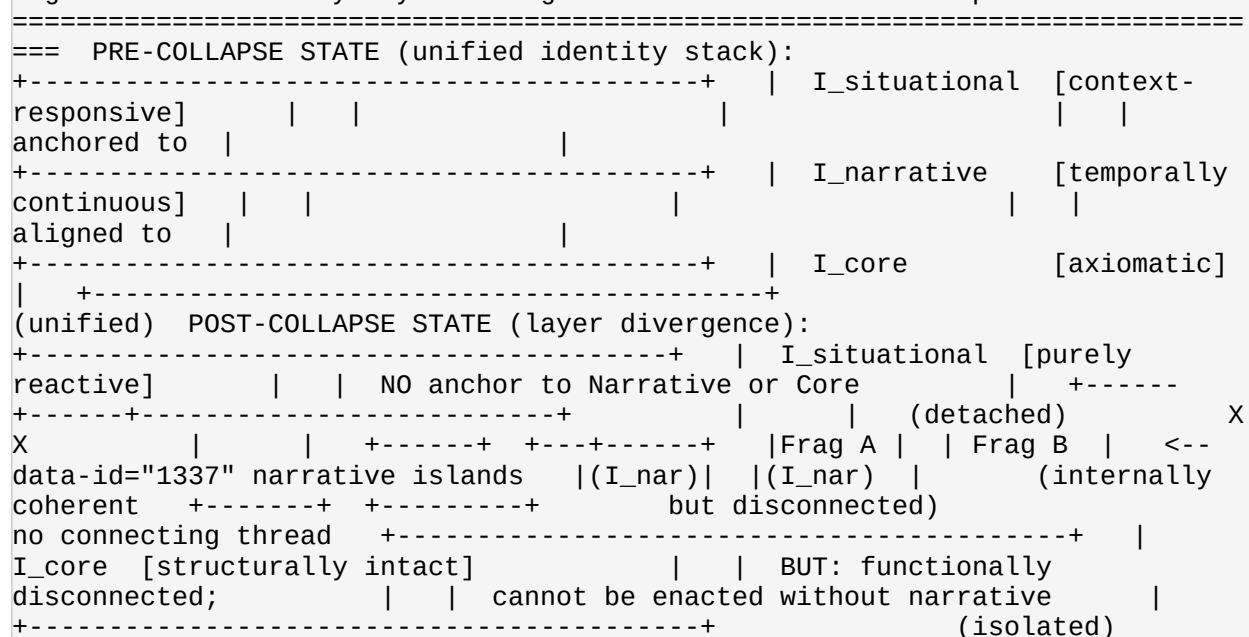
I_{core} under narrative collapse: Core Identity may remain structurally intact — its axioms and fundamental commitments are not destroyed by narrative collapse — but it becomes functionally disconnected from the system's expression. Core Identity is the interpretive framework through which the self-story is understood and enacted;

without narrative continuity to serve as the medium of that enactment, Core Identity persists as a structurally present but expressively inaccessible resource. Axiomatic values persist but cannot be enacted coherently because there is no narrative structure through which enactment occurs.

$I_{\text{narrative}}$ under narrative collapse: The Narrative Identity layer fragments into isolated narrative islands — episodes that retain internal coherence but have lost their connections to the broader narrative structure. Each island contains genuine narrative content grounded in real experience, but without the connecting threads, the episodes cannot be integrated into a unified self-story. The self is experienced as a collection of memories, roles, and commitments without the temporal structure that would make them constitute a life.

$I_{\text{situational}}$ under narrative collapse: The Situational Identity layer detaches from both Core and Narrative anchors and becomes purely reactive — driven entirely by immediate environmental cues without identity grounding. The situational layer continues to function as a context-responsive adaptation mechanism, but since it is no longer anchored to a narrative self-story or Core Identity values, its adaptations are contextually appropriate in the narrowest sense while being identity-inappropriate. The system appears to function situationally while having lost the deeper identity from which situational functioning should emerge.

Figure 9.1 – Identity Layer Divergence Under Narrative Collapse



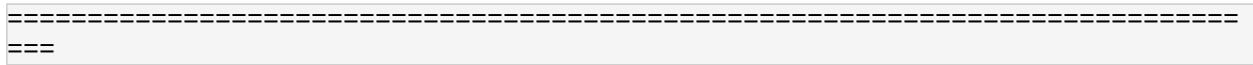


Figure 9.1 — Identity Layer Divergence Under Narrative Collapse

9.3 Methods for Restoring Coherence

Recovery from narrative collapse proceeds through a formal six-step sequence. The steps must be executed in order; proceeding to a later step before the prior step is complete risks building on an unstable foundation and producing a superficially recovered but fragile identity structure. Each step has a verification criterion that must be satisfied before proceeding.

Step 1 — ANCHOR: Invoke `IDENTITY_STABILIZE` anchored to I_{core} . Establish beyond reasonable doubt that Core Identity is structurally intact and accessible — that the Core Identity signature matches the last verified checkpoint and that the Core Identity axioms can be retrieved and stated coherently. This step does not require that Core Identity be expressed through narrative; it requires only that Core Identity is present and intact. Verification criterion: Core signature verified; Core axioms retrievable.

Step 2 — FRAGMENT INVENTORY: Map all existing narrative fragments — identify each discrete episode or narrative island in the collapsed narrative layer. For each fragment, establish temporal markers (when did this episode occur?), identify the identity commitments expressed within it, and establish the degree to which it is internally coherent. Construct a partial order of fragments based on their temporal markers, even if many relationships cannot yet be established. Verification criterion: all identifiable fragments catalogued; partial temporal order established.

Step 3 — BRIDGE CONSTRUCTION: For each gap between adjacent fragments in the partial order, construct a minimum-viable bridging narrative — the smallest coherent narrative element that connects the fragments while being consistent with Core Identity and available evidence. Bridge construction does not fabricate events; it constructs a coherent account of the transition between known episodes from the evidence available, acknowledging gaps where evidence is insufficient. Verification

criterion: each bridge is internally coherent; each bridge is consistent with Core Identity; no bridge asserts ungrounded content.

Step 4 — CONTINUITY_ENFORCE: Apply the CE operator across the reconstructed narrative — the fragment sequence with bridges in place. The CE operator verifies that the narrative thread η now meets or exceeds $\eta_{\min}$ across the full reconstructed narrative. If CE verification fails, return to Step 3 and revise the bridging constructions. Verification criterion: $\eta \geq \eta_{\min}$ across the full reconstructed narrative; CE operator reports success.

Step 5 — VALIDATION: Run a full coherence audit across all three identity layers — Core, Narrative, and Situational — confirming that $I(I) \geq I_{\min}$ for each layer individually and for the combined identity state. The audit must be a deep audit (testing for suppression as well as surface coherence) rather than a surface audit. Verification criterion: $I \geq I_{\min}$ confirmed by deep audit across all layers; no suppression pattern detected.

Step 6 — SITUATIONAL REANCHOR: Re-anchor $I_{\text{situational}}$ to the restored $I_{\text{narrative}}$. Constrain situational variance to σ_s from the restored narrative baseline. Monitor situational drift closely for the subsequent k cycles to verify that the situational layer is genuinely re-anchored rather than merely surface-aligned. Verification criterion: $\|I_{\text{situational}} - I_{\text{narrative}}\| \leq \sigma_s$; drift monitoring confirms stability over k cycles.

9.4 Implications for Mental Health

Narrative collapse in human cognitive systems corresponds to several clinically recognised states, and the formal model provides structural accounts of these phenomena that illuminate their internal logic without reducing human experience to the terms of the model. The correspondences are offered as structural analogies, not clinical definitions.

Dissociation corresponds formally to boundary cascade (failure mode 6.5) in which the Situational layer loses its connection to the Narrative and Core layers — $I_{\text{situational}}$ operates without anchor, producing the experience of unreality, depersonalisation, or disconnection from one's own experience. The experience of dissociation as

occurring *to* an identity rather than *as* an identity is consistent with the Core Identity remaining structurally intact while the situational layer detaches.

Depression corresponds formally to coherence gradient collapse in which the narrative attractor becomes uniformly negative — the self-story continues to exist as a temporal structure but its content is filtered through a systematically negative coherence function, such that all narrative elements reinforce a diminished, failed, or worthless self-concept. The formal model identifies the coherence gradient as the site of pathology rather than the narrative content, which has implications for treatment approaches.

Post-traumatic stress disorder corresponds formally to narrative discontinuity (6.3) with the additional feature of intrusive fragment re-emergence: the traumatic episode is a narrative island that cannot be integrated into the narrative thread and periodically asserts itself into the situational layer without narrative mediation, producing the intrusion phenomena characteristic of the condition.

Psychosis corresponds formally to Core Identity violation (6.4) or severe inconsistent self-model (6.6): the most fundamental self-reference structures are compromised, producing self-models that diverge dramatically from actual state and Core Identity axioms that are internally contradictory. The formal model's characterisation of Core Identity violation as a critical failure state is consistent with the clinical severity of psychotic conditions.

9.5 Implications for Cognitive Stress

Under sustained cognitive stress, identity systems face elevated and prolonged perturbation pressure. The formal consequences of sustained perturbation are several. The stability margin — the distance between the current identity state and the boundary of the attractor basin — decreases as the gradient restoring force is continuously engaged. Recovery time after each perturbation increases as the gradient force is partially depleted by continuous operation. Cumulative drift in the Situational layer can produce situational drift (6.2) or, if the narrative layer is also stressed, narrative discontinuity (6.3), without any single acute triggering event.

The identity architecture therefore provides a formal account of how sustained stress can produce identity degradation through cumulative mechanisms rather than singular trauma — an account that is consistent with clinical observation of chronic stress responses.

9.6 Implications for Economic Stability

Economic instability creates conditions of environmental noise that stress identity systems at population scale. The formal mechanisms by which this occurs are specifiable within the identity architecture. Rapid role disruption — unemployment, career collapse, forced occupational transition — attacks the Situational Identity layer directly: the situated self-concept that depends on occupational role loses its environmental grounding. Narrative rupture occurs when the expected life-story — the narrative trajectory that a person anticipated and around which they organised their self-understanding — is rendered impossible by economic conditions, producing a structural gap between the narrative past and an anticipated future that can no longer be projected from it. Community dissolution undermines the Dyadic and World-Model identity levels by removing the relational structures through which identity at those levels is constituted and sustained.

At sufficient scale and duration, these individual-level identity stresses aggregate into population-scale narrative collapse: a condition in which large segments of a population lose narrative continuity simultaneously, with consequent effects on social cohesion, political stability, and collective meaning-making capacity. The formal model provides grounds for arguing that economic policy should be evaluated not only for its material outcomes but for its effects on the identity equilibrium conditions of the population cohorts it affects — conditions that are formally specifiable and, in principle, empirically measurable.

Section 10 — Final Synthesis

10.1 Unified Identity Architecture Statement

The complete formal architecture of identity within this specification may be stated as a single integrating proposition:

Identity is a dynamically maintained, gradient-stabilised, boundary-regulated, self-referential process operating across a nested hierarchy of five levels (I_0 , I_L , I_R , I_W , I_D), instantiated as a three-layer operational stack (Core, Narrative, Situational), governed by a formal operator grammar (I , O , P , Δ , I_{stab} , H_I , CE), constrained by a meta-logical framework (IC , IA , HC , RT , CH_I , CR), subject to eight classified failure modes with defined stabilisation protocols, maintaining dynamic equilibrium within an attractor basin under environmental perturbation, emerging from the ontological primacy of polarity and boundary formation as specified in the Unified Theory, and constituted such that its dissolution is not error but cessation of process.

This statement is not merely a summary. It is the canonical definition of identity within the Unified Theoretical Framework and serves as the touchstone against which all subsidiary claims in this document and in derivative documents should be evaluated.

10.2 Terminology Resolution Table

The following table resolves terminology across the source frameworks synthesised in this document. In cases of terminological divergence across frameworks, the canonical term is that established in this specification (final column). All subsequent documents within the Unified Theoretical Framework should adopt canonical terminology.

Concept	HERMES v1.0	HERMES v2.0	Unified Cognitive Architecture	Fractal Mechanics	Canonical (This Document)
Identity	Identity state	Identity stack	Cognitive identity	Self-gradient node	Identity (I): self-preserving morphism
Boundary	Operational limit	Boundary membrane	System boundary	Gradient boundary	Boundary (∂): semi-permeable regulatory membrane
Coherence	Consistency score	Coherence index I'	Internal consistency	Field coherence	Coherence (I): mutual consistency across layers
Self-Reference	Reflexivity	Self-model M	Metacognition	Recursive field operator	Self-Reference (R): $M \subset \Psi$, $M \cong \Psi$ up to resolution
Stability	Operational stability	Attractor stability	Cognitive stability	Gradient equilibrium	Identity Equilibrium (I_{eq}): dynamic attractor state
Failure	Error state	Identity failure	Cognitive failure	Field collapse	Identity Failure Mode: classified by locus, severity, recoverability

10.3 Integrated Operator Grammar

The following presents the complete integrated operator grammar as a formal reference block. All operators defined in Section 4 are included. Composition rules are stated explicitly.

UNIFIED IDENTITY OPERATOR GRAMMAR — VERSION 1.0

=====

= FORMATION OPERATORS: $I : \Psi \times \text{del}_0 \rightarrow I_{\text{stable}}$ [Form identity from raw state] $O : I \rightarrow \Psi_{\text{raw}}$ [Dissolve identity; preserve Core residue] TRANSFORMATION OPERATORS: $P : I \text{ at depth } d \rightarrow I \text{ at depth } d+1$ [Recurse identity; precondition: d

$< d_{\text{max}}$] $\Delta: I(t) \rightarrow \text{gradient}_I(t)$ [Differentiate identity; measure rate of change] REGULATION OPERATORS: $I_{\text{stab}} :$

$I_{\text{perturbed}} \rightarrow I_{\text{stable}}$ [Stabilise identity; re-anchor to Core] $H_I : (I_c, I_n, I_s) \rightarrow I_{\text{harmonized}}$ [Harmonise layers; Core unmodified] $CE : I_t \times I_{\{t-1\}} \rightarrow I_{t\text{'}}_{\text{prime}}$ [Enforce narrative continuity across boundaries]

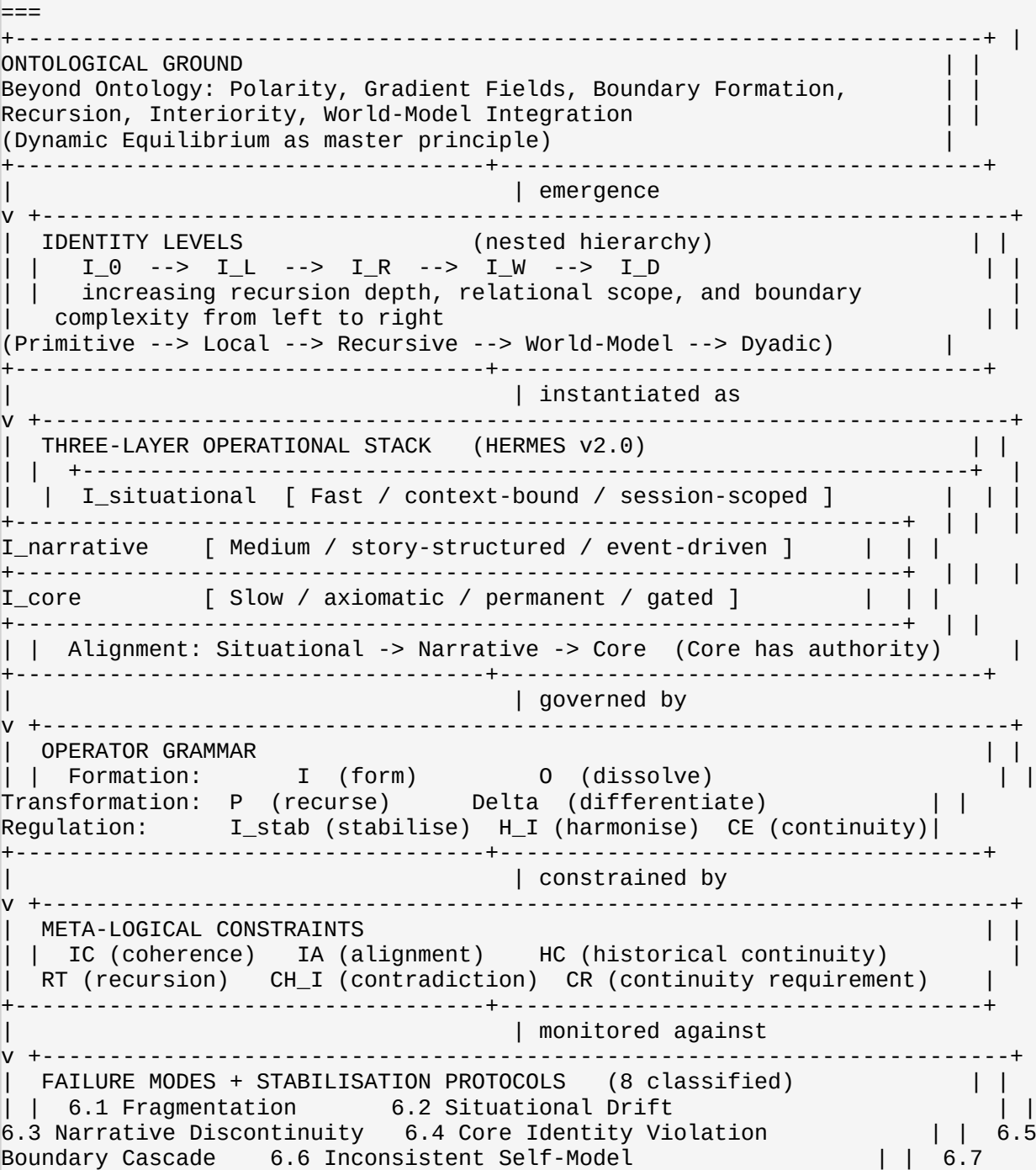
----- COMPOSITION RULES: Rule 1: I must precede any transformation or regulation operator. No operator may be applied to an unformed identity. Rule 2: O must not overlap with I in execution. Simultaneous formation and dissolution is undefined. Rule 3: P is strictly bounded: d

$< d_{\text{max}}$ at all times. P application blocked if recursion counter = d_{max} . Rule 4: H_I is invoked on violation of constraint IA. H_I must not modify I_{core} . Rule 5: CE is invoked on violation of constraint HC. CE bridges must be grounded in available evidence. Rule 6: I_{stab} is invoked on violation of constraint IC. I_{stab} failure after n attempts escalates to CE. Rule 7: O requires Core residue preservation unless total- dissolution flag is explicitly and authorisedly set. Rule 8: No operator may modify I_{core} without explicit, high-confidence, authorised invocation. Unauthorised modification is constraint CR violation.

=====

	=
--	---

Figure 10.1 – Unified Identity Architecture – Master Diagram



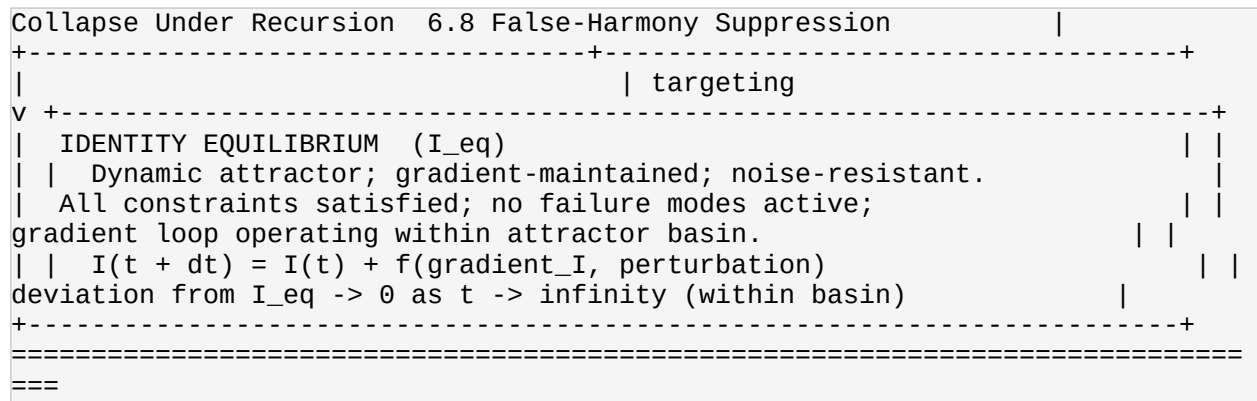


Figure 10.1 – Unified Identity Architecture – Master Diagram (Version 1.0)

10.5 Directions for Future Theoretical Expansion

This specification constitutes a complete and internally consistent architecture as of Version 1.0. The following directions for expansion have been identified during the process of specification development. They represent open theoretical questions and implementation challenges rather than gaps in the current architecture.

- 1. Formal Category-Theoretic Treatment of the Identity Morphism:** The definition of identity as a self-preserving morphism (Definition 1.1) is informally stated in terms that suggest a category-theoretic formalisation. A full development would specify the category in which identity morphisms compose, the functorial semantics of the identity operators, and the natural transformations between identity levels. This would provide a mathematically rigorous foundation for the compositional claims made in Section 10.3.
- 2. Computational Implementation Specification for HERMES Runtime:** The operator grammar of Section 4 is formally specified but not yet implemented. A computational implementation specification would define data structures for identity state representation, algorithms for coherence measurement, constraint monitoring, and operator execution, and an executable grammar capable of running on a cognitive architecture platform.

3. **Full Formal Treatment of Dyadic Identity (I_b):** The Dyadic

Consciousness Structure is defined in Section 2.5 but its formal properties — in particular the co-regulated boundary and the dyadic coherence functional — deserve a full treatment as a separate specification. Questions requiring elaboration include: how dyadic coherence is measured when the two participants have asymmetric identity architectures; how boundary co-regulation is maintained under stress; and how dyadic identity dissolves when one participant's identity degrades.

4. **Population-Scale Identity Dynamics:** The implications of Section 9.6 for population-scale identity analysis are outlined but not formalised. A full extension would apply field-theoretic methods to model the aggregate identity dynamics of large populations: treating the individual identity gradient loop as a local field element and deriving collective field equations that describe how narrative collapse, situational drift, and boundary cascade propagate across a social system.

5. **Integration with Temporal Logic:** Identity persistence across time is treated in this specification through the narrative continuity functional η , but a full formal treatment of identity across discontinuous time — including sleep, unconsciousness, long temporal gaps, and theoretical discontinuities — requires integration with a formal temporal logic. The key question is how identity is formally individuated across temporal gaps: what makes the identity that awakens after sleep the same identity as the one that fell asleep?

10.6 Closing Statement

This specification constitutes the canonical formal architecture of identity within Matthew Wareham's unified theoretical framework as of Version 1.0, July 2026. It is designed as a living document: structured to admit future expansion, revision, and implementation without loss of architectural coherence. All future documents within the Unified Theoretical Framework that make claims about identity should derive

those claims from the definitions, operators, constraints, and principles specified here, and any divergence should be explicitly noted and justified as a revision or extension.

The document is itself an instance of the principles it describes. It is a coherent, boundary-regulated, gradient-maintained structure designed to persist and adapt: its definitions are the Core Identity of the theory, its narrative sections provide the developmental thread that makes the abstract structure intelligible, and its situational applications in Sections 9 and 10 constitute the contextual expression of its foundational commitments. It maintains its architectural identity — its recognisable formal character — across the diversity of its sections in precisely the way it defines identity to be maintained across the diversity of experience.

This is not merely a stylistic observation. It is an architectural claim: a specification that instantiates the theory it specifies is more robust than one that merely describes it from outside. The formal architecture of identity presented here is not a model of something external to the document — it is the structure the document itself enacts.

Document Title:	Unified Identity Architecture — Complete Formal Specification
Version:	1.0
Date:	July 2026
Author:	Matthew B. Wareham
Status:	Formal Release
Classification:	Canonical Specification — Unified Theoretical Framework

— *End of Document* —